Single-Qubit Trajectories Workshop

Student Lab Handout

Session Overview

Duration: 1h15m
Format: Paired coding - predict, build, visualize 3 trajectories.

Session Objectives

- Build 3-5 gate circuits and plot step-by-step Bloch sphere trajectories.
- Predict paths on paper BEFORE coding (H=equator, S/T=phase, rotations=arcs).
- Hit precise Bloch targets using H-RY-RZ decomposition.
- Analyze 3 required trajectories + 1 bonus "gallery walk" circuit.

Timed Agenda:

- 0-10min: Setup
- 10-55min: 3 required tasks (15/15/20min)
- 55-70min: Gallery walk + bonus trajectory
- 70-75min: Submission

1. Setup

```
!pip install qiskit qiskit-aer matplotlib  # uncomment if needed
import numpy as np
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector
from qiskit.visualization import plot_bloch_multivector
from qiskit_aer import AerSimulator
%matplotlib inline
sim = AerSimulator()
print('Trajectory lab ready!')
```

2. Multi-State Trajectories (5min)

Key pattern: Build cumulative circuits, collect states, plot all together.

```
# Template - predict path FIRST!
# Cumulative qc.copy() pattern
qc0 = QuantumCircuit(1); state0 = Statevector(qc0)      # |0>
qc1 = qc0.copy(); qc1.h(0); state1 = Statevector(qc1) # H|0>
qc2 = qc1.copy(); qc2.s(0); state2 = Statevector(qc2) # S|+>

# Plot EACH state individually
plot_bloch_multivector(state0, title="Step 0: |0>")
plot_bloch_multivector(state1, title="Step 1: H|0>")
plot_bloch_multivector(state2, title="Step 2: S|+>")
print(" Cumulative qc.copy() pattern - use for ALL tasks!")
```

TASK 1: Equator Tour (15min)

H $\rightarrow$ **T** $\rightarrow$ **T** $\rightarrow$ **T** $\rightarrow$ **T** $\rightarrow$ **T** $\rightarrow$ **T** $\rightarrow$ **T** (8th roots of unity, 45° steps)
Predict: Full circle around equator? Clockwise/counterclockwise?

```
# TASK 1: Copy + complete 8 T gates
qc0 = QuantumCircuit(1); state0 = Statevector(qc0)      # |0>: (0,0,1)
qc1 = qc0.copy(); qc1.h(0); state1 = Statevector(qc1) # |+>: (1,0,0)
...
```

Step	Bloch (x, y, z)	Angle from prev	Gate effect
$\|0\rangle$	(0,0,1)	–	Start
H	(1,0,0)	90°	Equator
T	-	-	-
T	-	-	-
...	...	...	...
Total path	_____°	_____	

TASK 2: Pauli Ladder (15min)

$|0\rangle \rightarrow X \rightarrow Y \rightarrow Z \rightarrow X$ (pole hopping)

Predict: Final position relative to start?

```
# TASK 2: Complete Pauli sequence
```

Step	Bloch (x, y, z)	Angle from prev	Gate effect
$\|0\rangle$	(0,0,1)	–	Start
-	-	-	-
...	...	...	...
Total path	_____°	_____	

TASK 3: Precision Target (20min)

Challenge: Construct a single-qubit circuit that maps $|0\rangle$ to the exact Bloch vector $(0.6, 0.5, 0.6)$.

Requirements:

Normalize the target vector: $\vec{r} = [0.6, 0.5, 0.6]$

Derive the polar angle θ and azimuthal angle ϕ:

$$\theta = ? \quad \text{(from } z\text{-component)} \quad \phi \qquad\qquad = ? \quad \text{(from } x, y\text{-components)}$$

Build circuit:

Verify: Bloch vector error $< 10^{-10}$

```
# TASK 3: Hit target <0.01 error
target = np.array([0.6, 0.5, 0.6])
```

BONUS: Gallery Walk Circuit (10min)

Build your most interesting path (5 gates max):

```
# BONUS: Your creative trajectory
qc_best = QuantumCircuit(1)
# Add 5 gates max
```

Step	Bloch (x, y, z)	Angle from prev	Gate effect
$\|0\rangle$	(0,0,1)	–	Start
-	-	-	-
...	...	...	...
Total path	_____°	_____	

Quantum Gates Reference

Gate	Bloch Effect	Angle	Example
H	North $\rightarrow$+X equator	90°	$\|0\rangle \rightarrow \|+\rangle$
X	North $\leftrightarrow$ South pole	180°	$\|0\rangle \rightarrow \|1\rangle$
S/T	Z-axis phase	90°/45°	Equator rotation
RX(θ)	X-axis arc	θ	Continuous rotation
RY(θ)	Y-axis arc	θ	Pole $\rightarrow$equator
RZ(θ)	Z-axis spin	θ	Phase only

Lab Submission Checklist

- ☐ 3 complete trajectories with plots
- ☐ All analysis tables filled (angles, insights)
- ☐ Target 3 error < 0.01
- ☐ Bonus "gallery walk" circuit

This lab = your single-qubit geometry toolkit for all future circuits!